

Particulate Matter Monitoring & Health Effects

Issue 2026-08-30 | Entry window 30 August 2026 (single day) | 14 records in scope, of which **3 are dated backfill**

14

RECORDS IN SCOPE
(17 SCREENED OUT)

7

SUBTOPIC CLUSTERS
OF TEN OCCUPIED

4

SENSING /
INSTRUMENTATION

4

EFFECT ESTIMATES
(ONE SOURCE)

7

RECORDS WITH NO
HEALTH ENDPOINT

Signal of the day

A city-scale low-cost sensor network is used as the *instrument of record* for a mesoscale meteorology question — and the calibration literature published alongside it says why that is still risky.

Chen et al. (*Environ Sci Technol*) composite **42 lake-breeze events** over Chicago in 2022 using one year of the Microsoft Eclipse low-cost sensor network plus meteorological stations. Weekday lake breezes *reduce* O_3 by up to ~ 6 ppb along the northeast shoreline — the opposite of the sea-breeze ozone exceedances reported for most coastal regions — while *raising* $PM_{2.5}$ by up to $\sim 3 \mu g m^{-3}$ through accumulation in a shallower boundary layer and enhanced semivolatile partitioning at lower temperature. A $3 \mu g m^{-3}$ signal is precisely the magnitude that the three calibration papers carried alongside it show is *recoverable but not free*: Blaga et al. report Pearson r moving from -0.28 to 0.89 in the humid season after a two-step temperature/RH correction, and from 0.06 to 0.65 in the dry season — and the uncorrected Bucharest network flagged exceedances at 8 of 44 sites where the corrected one flagged all 44. The methodological reading for distributed PM sensing is blunt: an LCS network is now good enough to resolve a few- $\mu g m^{-3}$ meteorological forcing, but only after a seasonally stratified, site-specific correction, and Senarathna et al. show a wet-season model applied to dry-season data carries **26.57% MAPE** and does not transfer 115 km between Colombo and Kandy.

What changed today

- **Calibration transferability is the binding constraint, stated three ways.** Blaga (Bucharest, 44 Clarity Node-S), Senarathna (Sri Lanka, TSI BlueSky, two climatic zones) and Sengupta (Oxford OxAria, LSTM) independently report the same failure mode: a correction fitted at one site and season degrades elsewhere. Sengupta's contribution is that a sequence model reaches **9.1% expanded uncertainty for $PM_{2.5}$** against the Equivalence Spreadsheet Tool — inside the EU indicative-monitoring threshold — by learning time-lagged environmental dependence a Random Forest treating observations independently cannot.
- **The regulatory consequence of *not* calibrating is now quantified.** Bucharest's manufacturer pre-calibration understated $PM_{2.5}$ enough that only 18% of sensors appeared to breach the EU $25 \mu g m^{-3}$ daily limit; after correction every site did, averaging >60 exceedance days against 35 permitted. This is the strongest argument in the day's harvest for treating uncalibrated LCS output as unfit for compliance framing.
- **An operational null worth keeping.** Zhang & Yu's Wuhan time-series (906 days, 391,344 ED visits) recovers textbook associations — $PM_{2.5}$ AQI RR **1.028 (1.015–1.042)** per IQR at lag 0–2, PM_{10} 1.030, NO_2 1.049 — and then shows that *adding those variables did not improve* day-ahead demand forecasting (MAE 33.2 with meteorology, unchanged with pollution). Statistical detectability and operational value are dissociable, and papers rarely test the second.
- **Detection method, not environment, dominated airborne-microplastic heterogeneity.** Rahmani et al. pool 27 studies across 16 countries at $I^2 > 99\%$ and find the largest moderator is *instrument* — FTIR consistently reporting higher counts than microscopy or Raman. Their indoor/outdoor contrast ($1,380$ vs. $102 MP m^{-3}$) should not be read as an exposure ratio until sampling and detection protocols are harmonised, and they say so.
- **Two records ship on metadata alone.** The *Aerosol Air Qual Res* oil-mist ventilation study and the *Environ Int* solid-fuel-PM nitrogen-organics paper deposited no abstract to Crossref and are absent from Europe PMC, PubMed and Semantic Scholar. They are listed, not summarised, and carry no extracted numbers.
- **Health-side yield was low and skewed to synthesis.** Of 7 PubMed records entering on 30 August, one was off-topic (flotation process water); the remainder are two reviews, one animal study, one epigenetics cohort where air pollution is 1 of 25 mixture components, one composition campaign, and one cross-sectional performance study. No cardiovascular, respiratory or neuro-epidemiological record entered the window at all.

Corpus analytics

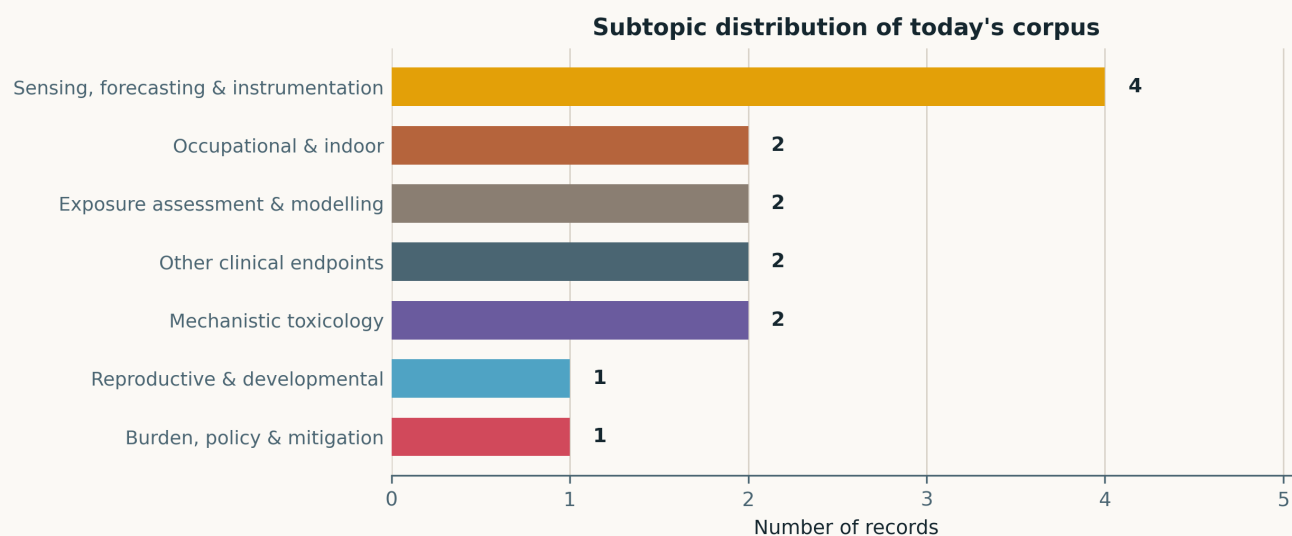


Fig. Subtopic assignment of the 14 in-scope records. Seven of the ten clusters are occupied; *Cardiovascular & metabolic*, *Neuro / mental health* and *Respiratory & allergic* are all empty. Checked against `metrics.csv`: this is the fourth issue this month with an empty respiratory cluster (21, 23 and 28 August were the others), but the *simultaneous* absence of respiratory, cardiovascular and neuro work is unique in August. *Sensing, forecasting & instrumentation* leads at 4, but three of those four are dated backfill recovered through Consensus and Crossref rather than deposits entering PubMed on 30 August — do not read this as a one-day flux.

Study architecture

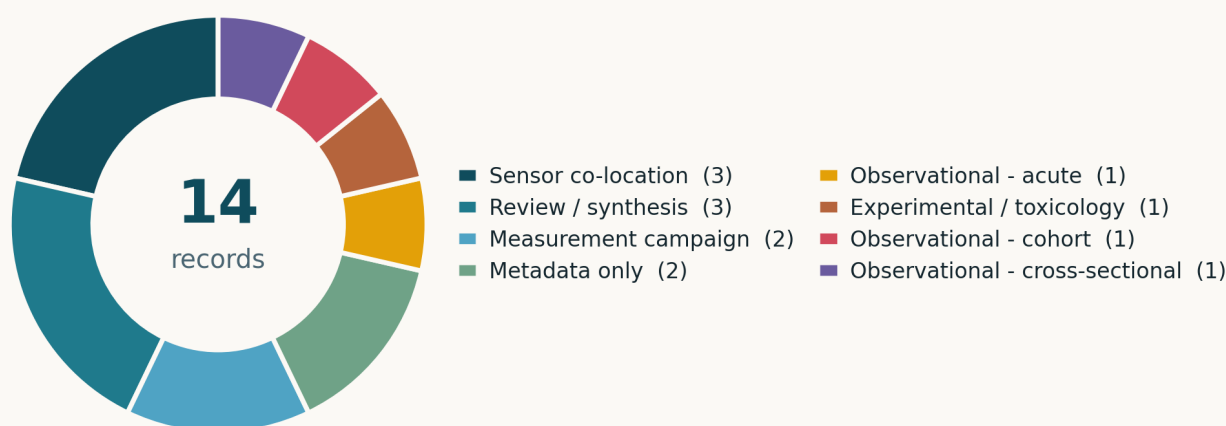


Fig. Study architecture. Sensor co-location is the single most common design (3), followed by measurement campaigns (2) and reviews/syntheses (3, counting the microplastics meta-analysis, the ILC2 review and the Southeast Asia scoping review). Two records are *Metadata only*: the architecture is genuinely unknown, not unmapped.

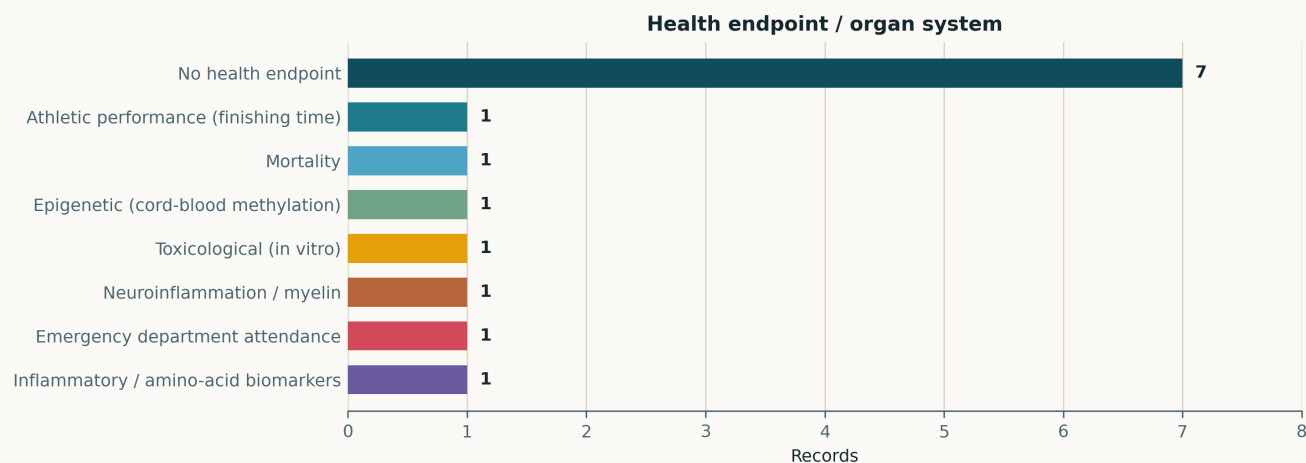


Fig. Health endpoint by organ system. Seven of fourteen records carry no human health endpoint — four sensing/calibration records, two composition or exposure-assessment studies, and the metadata-only oil-mist record. Of the seven that do, no two share an endpoint: emergency-department attendance, mortality, athletic performance, neuroinflammation, cord-blood methylation, in-vitro toxicology and inflammatory biomarkers each appear once. That dispersion is why no pooled health estimate is attempted below.

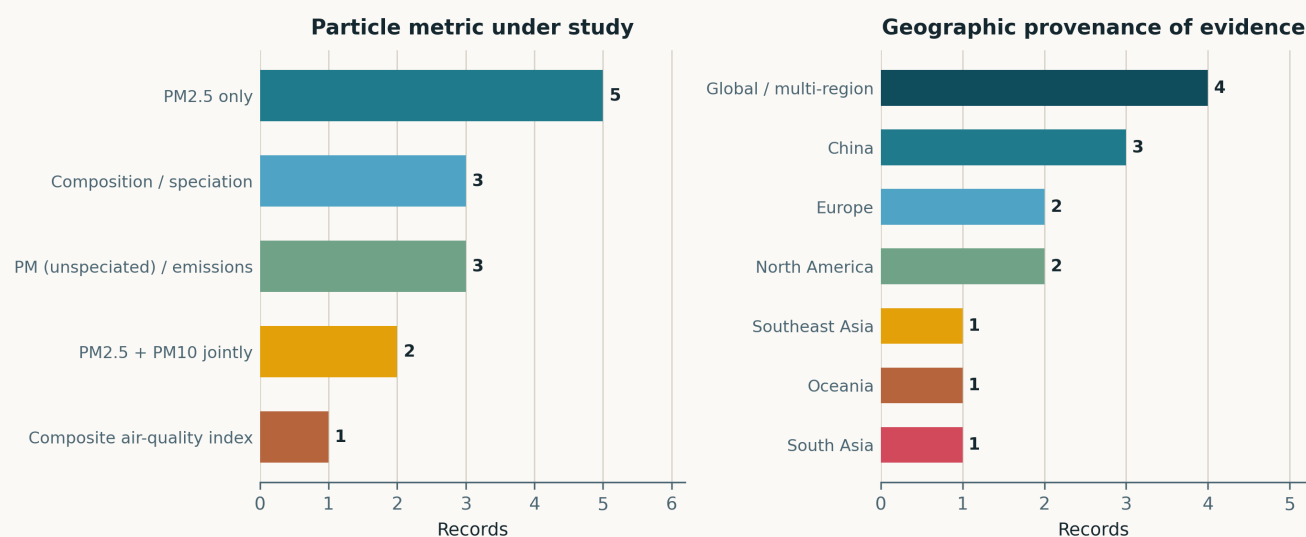


Fig. Left — particle metric. PM_{2.5} alone is the exposure in 5 records; a further 3 report composition or speciation and 3 use unspiciated PM or emissions. Ultrafine particles are again absent. Right — geographic provenance. China contributes 3; the single South Asian record (Sri Lanka, a calibration study) and the single Southeast Asian record (the attribution review) are both present. Africa is absent today, but this is a one-day gap and not a trend: the 29 August issue carried two African records (South Africa; a multi-country sub-Saharan analysis).

Life-course windows addressed today (records may span several stages)

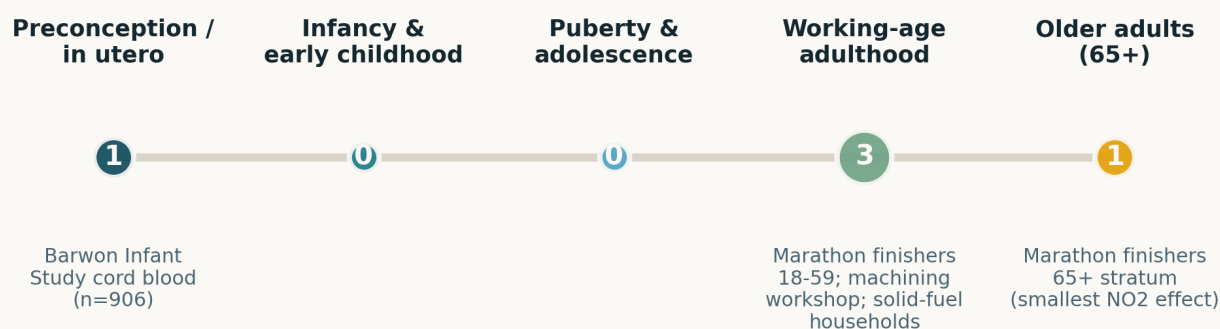


Fig. Life-course windows addressed. Counts are non-exclusive and only five records carry an identifiable window at all. The in-utero cell is the Barwon Infant Study cord-blood analysis; the working-age cell holds the marathon cohort's 18-59 strata, the machining workshop and the solid-fuel households; the 65+ cell is the marathon cohort's oldest stratum, in which the NO₂ effect was smallest (1.32 vs. 2.17 min per $\mu\text{g m}^{-3}$ in 18-29-year-olds). Infancy/early childhood and adolescence are empty.

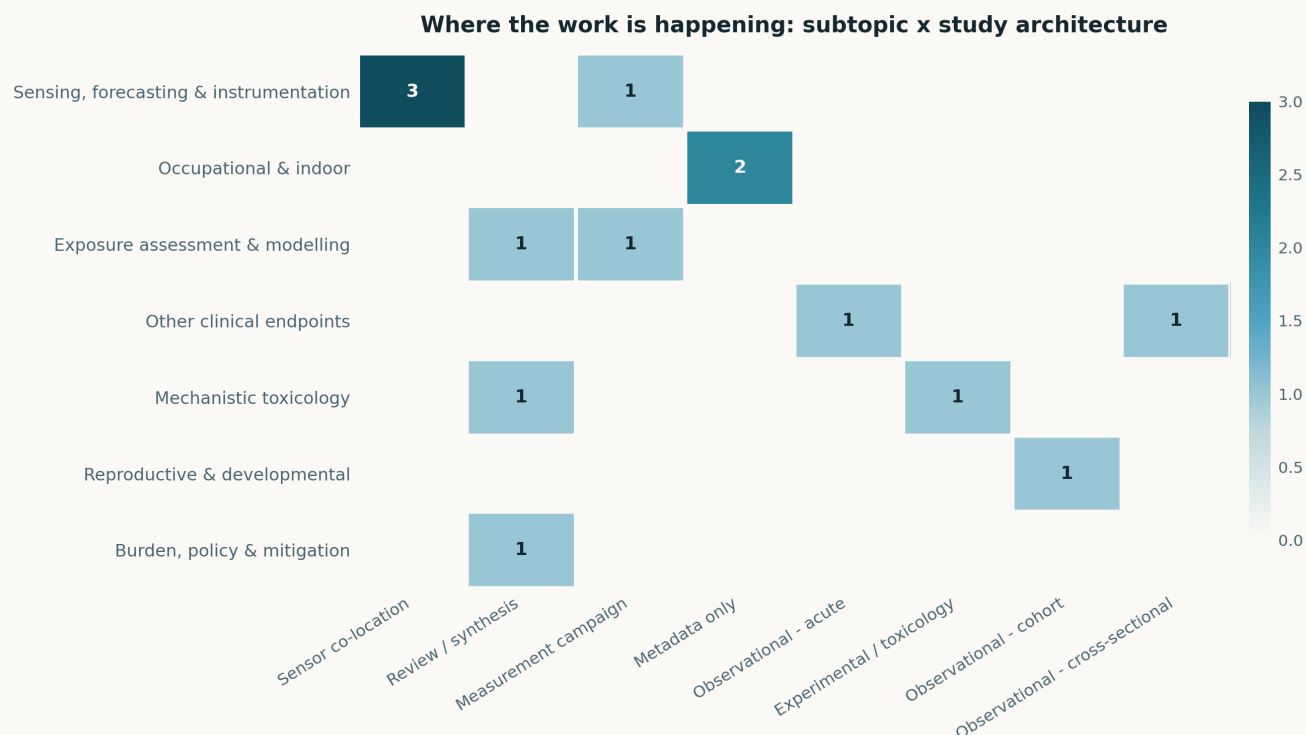


Fig. Subtopic × study-architecture cross-tabulation. The structural point is the same one this digest has recorded repeatedly: the *Sensing* row is entirely co-location and campaign work with no health-endpoint linkage, and the health rows contain no instrumentation. Nothing in today's harvest bridges the two.

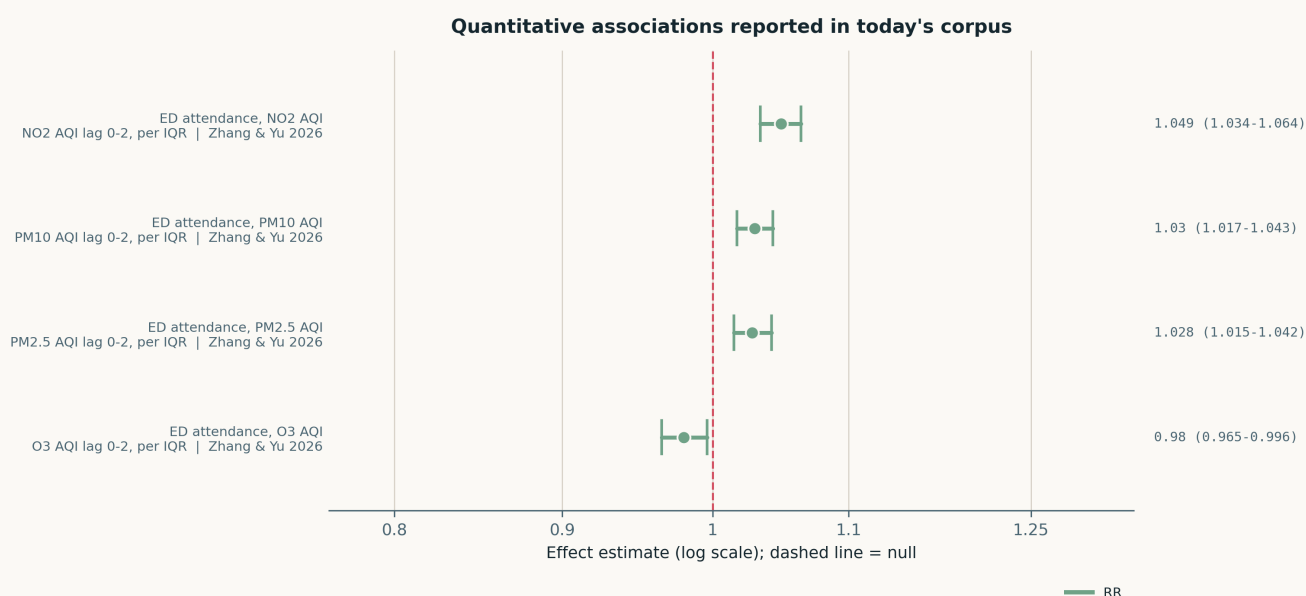


Fig. All four ratio-metric estimates with confidence intervals in the corpus. They come from a single unrefereed preprint (Zhang & Yu, Research Square) and are mutually adjusted co-pollutant estimates from one model, so the plot shows within-study precision and direction only — it is not a synthesis and carries no heterogeneity statistic. The O₃ estimate is protective (0.980, 0.965–0.996), which in a Wuhan time-series is most plausibly seasonal confounding rather than a causal effect. Donaldson et al.'s marathon estimates are absolute time differences in minutes and are deliberately excluded: on a log-ratio axis they would misread as null.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked [backfill] were recovered on 30 August but were published earlier in 2026; they are dated to this issue and must not be read as same-day deposits.

Sensing, forecasting & instrumentation

4 records

Chen et al. 2026k Environ Sci Technol

Lake-breeze effects on urban air quality resolved by a city-wide low-cost sensor network

One year (2022) of the Microsoft Eclipse LCS network plus meteorological stations over Chicago, composited into **42 lake-breeze events** and 51 weak events confined within a few kilometres of the shore. Weekday events reduce O_3 by up to ~ 6 ppb along the northeast shoreline — attributed to NO titration in a shallow boundary layer, inflow of cleaner lake air, and suppressed photochemistry at lower temperature — while *raising* $PM_{2.5}$ by up to $\sim 3 \mu g m^{-3}$ via accumulation of local emissions and semivolatile partitioning. The limitation that matters: event-composited means over an LCS network inherit that network's calibration bias, and no reference-monitor validation of the $PM_{2.5}$ increment is reported, so the $3 \mu g m^{-3}$ figure is a network-relative anomaly rather than a traceable concentration. Directly relevant precedent for using a dense LCS deployment as the primary instrument in a mesoscale study.

doi: 10.1021/acs.est.6c09290

Blaga et al. 2026 [backfill] PLOS Clim

City-scale calibration of a 44-node $PM_{2.5}$ network for regulatory-compliant assessment

44 Clarity Node-S sensors across Bucharest against National Environmental Protection Agency reference stations; nine co-located spatial clusters, one used to fit and eight held out. Manufacturer pre-calibration **systematically underestimated** $PM_{2.5}$, worst in summer when secondary aerosol formation dominates. A two-step seasonal model on temperature and relative humidity alone moved Pearson r from **0.06 to 0.65** (dry season) and from **-0.28 to 0.89** (humid season). Applied network-wide, calibrated sensors average **>60 days/yr** above the EU $25 \mu g m^{-3}$ daily limit against 35 permitted; pre-calibration, only 8 of 44 (18%) exceeded. Weakness: calibration degrades in hot dry low-concentration conditions and inter-sensor variability in calibration performance remains large, so a single network-wide model is not defensible. The 1:8 train/validate split is the design detail worth copying.

doi: 10.1371/journal.pclm.0000780

Senarathna et al. 2026 [backfill] Environ Monit Assess

Seasonal and spatial non-transferability of LCS calibration across two tropical climatic zones

TSI BlueSky sensors in a harmonised network across Colombo (wet-zone) and Kandy, compared under linear regression, multiple linear regression and an echo-state network. Harmonised sensors agreed within **10%** of one another. Performance improves monotonically with averaging window: at 24 h, wet season $R^2 = 0.60$, MAE $0.93 \mu g m^{-3}$, MAPE 7.81%; dry season $R^2 = 0.85$, MAE 1.98, MAPE 9.17%. The two transfer results are the point — applying a wet-season model to dry-season data yields **26.57% MAPE**, and a Colombo-fitted model degrades in Kandy, while a Kandy-specific model with temperature and RH reaches $R^2 = 0.84$ – 0.92 . Limitation: no independent reference co-location is reported for the second zone at the same duration, so the spatial-transfer claim rests partly on inter-sensor agreement rather than reference truth.

doi: 10.1007/s10661-026-15623-4

Sengupta et al. 2026 [backfill] arXiv (preprint)

LSTM calibration of low-cost PM and NO_2 sensors against a regulatory equivalence test

Sequence model trained on co-located OxAria reference data in Oxford for $PM_{2.5}$, PM_{10} and NO_2 , benchmarked against a Random Forest baseline. The framing argument is that RF treats each hourly observation independently and therefore cannot represent sensor lag or delayed environmental response; the LSTM with time-lagged features, harmonic time encodings and interaction terms beats it on R^2 across train, validation and test. The result that carries weight is the external check: validation against Equivalence Spreadsheet Tool 3.1 gives expanded uncertainties of **9.1% ($PM_{2.5}$)**, **12.42% (PM_{10})** and **22.11% (NO_2)** — i.e. PM meets, and NO_2 misses, the indicative-monitoring bar. Unrefereed, single network, and no drift analysis over a multi-year horizon, which is where sequence models are most likely to fail.

doi: 10.48550/arXiv.2604.21527

Exposure assessment & modelling

2 records

Rahmani et al. 2026 Rev Environ Health

Airborne microplastics indoors and outdoors: pooled concentrations and the instrument problem

PRISMA systematic review to October 2023: 27 eligible studies across 16 countries, 21 pooled under a random-effects model. Indoor concentrations exceed outdoor — moderately polluted indoor environments $1,380.28$ vs. $102.45 MP m^{-3}$ outdoors — with indoor healthcare settings and outdoor urban areas the most contaminated sub-environments. The authors present the pooled figures as a descriptive summary rather than an estimate, correctly: heterogeneity is **$I^2 > 99\%$** and the dominant moderator is **detection instrument**, with FTIR reporting consistently higher counts than microscopy or Raman spectroscopy; sampling duration also matters. Sensitivity analysis excludes single-outlier dominance. For instrumentation work this is a clean external demonstration that method-driven variance can exceed environmental variance, which is exactly the risk in heterogeneous LCS fleets.

doi: 10.1515/reveh-2025-0059

Ren et al. 2026 iScience

Vertical contrast in $PM_{2.5}$ composition and secondary formation, Shiyuan, central China

One year across three urban sites and one mountain site. Annual mountain $PM_{2.5}$ is **70% of urban** levels, but sulfate–nitrate–ammonium is a *larger* fraction there (49.3%). SNA is vertically near-homogeneous in spring, summer and autumn (coefficient of divergence < 0.2) and stratifies only in winter, when most components are urban-enriched. Higher nitrogen oxidation ratio (0.27) and SOC/OC (0.65) at altitude indicate a more aged aerosol. Limitation: a single mountain site cannot separate altitude from the site's own upwind exposure, and no vertical profiling instrument is used — the "vertical distribution" claim rests on a two-point contrast. Relevant to siting logic for distributed networks: composition, not mass, is what varies with height here, and a mass-only node would report the mountain site as simply cleaner.

doi: 10.1016/j.isci.2026.117090

Occupational & indoor

2 records

Gu et al. 2026b Environ Int*Nitrogen-containing organics in household solid-fuel combustion PM vs. urinary inflammatory and amino-acid profiles*

Listed on Crossref metadata only. The deposit carries no abstract, and the record is absent from Europe PMC, PubMed and Semantic Scholar as of this run, so **no finding, effect size, cohort size or location is asserted here**. The title indicates an exploratory personal exposure design pairing speciated N-containing organic compounds in solid-fuel combustion PM with targeted urinary inflammatory and amino-acid metabolites — a personal-monitoring plus targeted-metabolomics pairing that is directly on-axis for this watch. Queued for the abstract retry list; it will be summarised, not re-listed, once text is available.

doi: 10.1016/j.envint.2026.110491

Li et al. 2026f Aerosol Air Qual Res*Non-uniform oil-mist distribution and local ventilation in a machining workshop*

Listed on Crossref metadata only — no abstract deposited, and no record in Europe PMC, PubMed or Semantic Scholar. **No quantitative claim is made here**. The title indicates spatial characterisation of oil-mist aerosol in a machining workshop with a local ventilation intervention, which is the occupational analogue of the indoor-siting problem: where a workshop's aerosol field is strongly non-uniform, a single fixed monitor is a poor estimator of personal exposure regardless of its accuracy. Queued for the abstract retry list.

doi: 10.1007/s44408-026-00161-y

Mechanistic toxicology**2 records****Jerome et al. 2026** J Immunotoxicol*World Trade Center dust: neuroinflammation and myelin response at 3 and 12 months*

Twelve-week-old male spontaneously hypertensive rats given intratracheal WTC dust at **2.43 ± 0.78 mg/day on two consecutive days**, then euthanised at 3 and 12 months. At 3 months, increased Black-Gold II myelin staining without change in myelin basic protein or CNPase; at 12 months, microglial activation (Iba1) and elevated brain **IL-1β**. The authors read the discrepant myelin markers as a compensatory response preceding degeneration. Three limitations bound the inference: a two-day bolus instillation is not a chronic inhalation exposure and bypasses the upper airway; the SHR strain is hypertensive by design, so the neuroinflammatory phenotype is confounded with a vascular one; and males only. Useful as a mechanistic anchor for the First-Responder cognitive literature, not as a dose-response.

doi: 10.1080/1547691X.2026.2722654

Luo et al. 2026b Clin Rev Allergy Immunol*Environmental pollutants and innate immunity via group 2 innate lymphoid cells*

Narrative review covering O₃, PM, diesel exhaust particles and cigarette smoke alongside nanomaterials and microplastics, organised around ILC2 biology. The proposed common route is epithelial barrier compromise followed by alarmin release — IL-33, IL-25, TSLP — with ILC2s as the sensor coupling environmental detection to type-2 inflammation, plus non-canonical regulation of ILC2 plasticity and functional adaptation. No new data, no quantitative synthesis, and PM is not separated from the other exposures, so nothing here constrains a PM-specific dose or particle-property relationship. Carried as a citation anchor for the epithelial-alarmin axis in respiratory PM work; tier C.

doi: 10.1007/s12016-026-09193-z

Reproductive & developmental**1 record****Tanner et al. 2026** Environ Epigenet*Prenatal exposure mixture and aromatase brain-promoter methylation in cord blood*

Barwon Infant Study birth cohort, **n = 906**, testing 25 prenatal exposures across five classes — manufactured chemicals, air pollution, pharmacological, nutritional and sunlight-related — against methylation of the brain-specific *CYP19A1* P1.f promoter using weighted quantile sum regression. The mixture index was associated with higher P1.f methylation, **adjusted mean difference 0.71 (95% CI 0.11–1.32)**, $p = 0.021$, interpreted as reduced brain aromatase activity. Bisphenols, reduced sunlight, household mould, phthalates, low folate and air pollution carried the largest weights. The inference-threatening limitation is structural: WQS assigns weights within a single index, so air pollution's individual contribution is neither estimated nor separable, and cord blood is a proxy for a brain-specific promoter. Tier C for this watch — PM is one component of a 25-exposure mixture.

doi: 10.1093/eep/dvag027

Burden, policy & mitigation**1 record****Simion et al. 2026** Lancet Reg Health West Pac*Climate-related health attribution in Southeast Asia: what is and is not being studied*

Scoping review, **5,294 records screened, 77 included**. Air pollution is the subject of **70% of studies**, linked to up to 31% of all-cause mortality and an estimated **296,000 annual deaths from wildfire smoke during El Niño events**; heat is second, associated with up to 8.61% of mortality in Bangkok. Floods, droughts, extreme rainfall, storms and sea-level rise are near-absent. The paper's own framing — that the research portfolio is misaligned with the regional hazard profile — is the transferable finding; the attributable fractions themselves are heterogeneously derived across 77 studies with no standardised attribution framework, which the authors name as the central methodological gap. For a monitoring programme the operative point is that the region's dominant PM source is episodic biomass burning, which mass-only fixed networks under-sample in both space and time.

doi: 10.1016/j.lanwpc.2026.101952

Other clinical endpoints**2 records**

Donaldson et al. 2026 Sports Med Open*Race-day air pollution and marathon finishing time across 2.76 million finishes*

2,756,553 net finishing times from six World Marathon Majors (Berlin, Boston, Chicago, London, New York, Tokyo), 75 marathon-year events, 2010–2024, with race-day NO₂ and PM_{2.5} from CAMS global reanalysis and linear mixed models adjusting heat index, wind and temporal trend. Each 1 µg m⁻³ NO₂ was associated with **+1.43 min (95% CI 0.66–2.21) in men and +1.56 min (0.63–2.48) in women**, with a steep gradient by ability — recreational 2.12 vs. elite 0.34 min per µg m⁻³, persisting as a percentage of mean time (0.67% vs. 0.18%) — and by age (2.17 min in 18–29-year-olds vs. 1.32 in ≥60). **PM_{2.5} was null** (–0.13 min, –0.67 to 0.40 in men). The exposure is reanalysis at city scale, not course-level or personal, and NO₂ at that resolution is a traffic-proximity proxy; the NO₂/PM_{2.5} divergence is therefore at least partly a measurement-error contrast rather than a toxicological one. Still the largest exercise-performance exposure dataset assembled.

doi: 10.1186/s40798-026-01099-6

Zhang & Yu 2026 (preprint) Res Sq (preprint)*Air pollution is detectable in ED demand and useless for forecasting it*

Single-centre daily time-series at Tongji Hospital Guanggu Branch, Wuhan, 1 Jan 2024 to 26 Jun 2026: **906 dates, 391,344 visits**, mean 431.9/day (SD 59.2), falling to 319.3/day over Spring Festival. Quasi-Poisson models give, per IQR of lag 0–2 pollutant-specific AQI, RR **1.028 (1.015–1.042)** for PM_{2.5}, 1.030 (1.017–1.043) for PM₁₀, 1.049 (1.034–1.064) for NO₂ and an inverse 0.980 (0.965–0.996) for O₃; each 1 °C is +0.85% visits. The contribution is the temporally-ordered validation: a history-plus-calendar model achieves MAE 33.6 visits/day, meteorology reduces it to 33.2, and **adding air-pollution AQI does not improve it at all**. Single centre, city-level AQI rather than catchment exposure, unrefereed. The inverse O₃ term is almost certainly seasonal confounding. Read as a warning that a statistically robust exposure–response can carry no incremental predictive information.

doi: 10.21203/rs.3.rs-10496137/v1

Corpus register

First author	Focus	Journal	Design	Metric	Region
Reproductive & developmental					
Tanner et al.	Prenatal environmental determinants of aromatase brain-promoter methylation in cord blood: chemical, airborne, pharmacological, and nutritional factors	<i>Environ Epigenet</i>	Birth cohort + weighted quantile sum regression	Air pollution as 1 of 25 prenatal exposures in a WQS mixture index	Australia
Mechanistic toxicology					
Jerome et al.	Prolonged brain cytokine elevation and microgliosis following exposure to World Trade Center dust particulate matter in male spontaneously hypertensive rats	<i>J Immunotoxicol</i>	Animal exposure study	WTC dust, intratracheal, 2.43 (+/-0.78) mg/day x 2 d	United States
Luo et al.	Impacts of Environmental Pollutants on Innate Immunity: The Role of Group 2 Innate Lymphoid Cells	<i>Clin Rev Allergy Immunol</i>	Narrative review	PM, diesel exhaust particles, O ₃ , cigarette smoke, nano-materials, microplastics	Multi-country
Occupational & indoor					
Gu et al.	Nitrogen-containing organic compounds in household solid-fuel combustion PM and residential urinary inflammatory and targeted amino-acid profiles: an exploratory personal exposure study	<i>Environ Int</i>	Metadata only	Personal PM from household solid-fuel combustion; N-containing organic speciation	Not stated (metadata only)
Li et al.	Study on the Non-uniform Distribution Characteristics of Oil Mist and Local Ventilation Method in Machining Workshop	<i>Aerosol Air Qual Res</i>	Metadata only	Oil-mist aerosol, spatial distribution	China
Sensing, forecasting & instrumentation					
Blaga et al.	City-scale calibration of a low-cost PM _{2.5} network for regulatory-compliant air-quality assessment	<i>PLOS Clim</i>	Sensor co-location + seasonal MLR calibration	PM _{2.5} , Clarity Node-S vs NEPA reference stations	Romania
Chen et al.	Lake Breeze Effects on the Urban Environment in Chicago	<i>Environ Sci Technol</i>	Sensor-network event analysis (lake-breeze compositing)	PM _{2.5} and O ₃ from the Microsoft Eclipse low-cost sensor network plus meteorological stations, 1 year	United States
Senarathna et al.	Enhancing low-cost PM _{2.5} air quality monitoring sensors through sensor calibration using regression and echo-state network models	<i>Environ Monit Assess</i>	Sensor co-location + regression / echo-state network calibration	PM _{2.5} , TSI BlueSky vs reference, wet and dry season	Sri Lanka
Sengupta et al.	A temporal deep learning framework for calibration of low-cost air quality sensors	<i>arXiv (preprint)</i>	Sensor co-location + LSTM sequence calibration	PM _{2.5} , PM ₁₀ and NO ₂ , LCS vs co-located reference (OxAria)	United Kingdom
Exposure assessment & modelling					
Rahmani et al.	Airborne microplastics: a comprehensive analysis of indoor and outdoor pollution patterns	<i>Rev Environ Health</i>	Systematic review + random-effects meta-analysis	Airborne microplastic number concentration (MP/m ³), indoor vs outdoor	Multi-country
Ren et al.	Contrasting chemical characteristics and secondary formation of PM _{2.5} from mountain and urban sites in central China	<i>iScience</i>	Source apportionment	PM _{2.5} mass and SNA / OC-EC speciation, annual	China
Burden, policy & mitigation					
Simion et al.	Attributing climate-related health burdens in Southeast Asia: a scoping review	<i>Lancet Reg Health West Pac</i>	Scoping review	Air pollution, heat and extreme-weather attribution studies	Southeast Asia
Other clinical endpoints					
Donaldson et al.	Air Pollution Impairs Marathon Performance: A Cross-Sectional Analysis of 2.7 Million Finishers Across Six World Marathon Majors	<i>Sports Med Open</i>	Observational - cross-sectional	Race-day NO ₂ and PM _{2.5} from CAMS global reanalysis, per 1 µg/m ³	Multi-country

First author	Focus	Journal	Design	Metric	Region
Zhang & Yu 2026	Environmental and calendar correlates of daily emergency department demand in Wuhan, China: a single-centre time-series study with temporal validation	Res Sq (preprint)	Time-series / case-crossover	Pollutant-specific AQI (PM _{2.5} , PM ₁₀ , NO ₂ , O ₃), lag 0-2, per IQR	China

Provenance and method

Window: 30 August 2026 alone, contiguous with the 29 August issue; no gap. Sources queried on 31 August 2026. **PubMed** [EDAT] 2026/08/30 — health axis 7 via the connector against 4 from the local harvester (the connector is again the better recall leg; the local set is a strict subset), instrumentation axis 0 on both. **Europe PMC** CREATION_DATE 2026-08-30 — 6 hits, of which 2 were already in the PubMed set and 2 were published in the 29 August issue, leaving 2 unique (one review, one Research Square preprint). **Crossref by-ISSN** across 25 tracked non-PubMed journals — 12 deposits with a 30 August created date, of which 3 are carried and 9 screened out as non-particulate (PFAS, PCB, soil vanadium, gas-phase ozone isotopes, aquatic pharmacology, biodiversity accounting). **arXiv** relevance query returned 60 entries, **none** published on or after 25 August. **OpenAlex** remains dead — HTTP 403 paid-plan block with intermittent 429 — and is routed through Consensus, which returned 10 hits on the calibration/drift axis: 7 resolved to DOIs already in `seen.json`, **3 were new** and are carried as explicitly labelled backfill (Blaga, Senarathna, Sengupta; publication dates February, June and April 2026). **31 unique records considered; 7 dropped as already seen; 14 in scope; 17 logged to rejected.json1 with a reason**. *Two records are listed on metadata only* — `10.1016/j.envint.2026.110491` and `10.1007/s44408-026-00161-y` deposited no abstract to Crossref and are absent from Europe PMC, PubMed and Semantic Scholar; they are not summarised and no numbers are attributed to them. Subtopic, design, particle-metric and geography labels are assigned from title, abstract, keywords and MeSH terms; assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. First-author surnames are read from PubMed `efetch` or the Crossref deposit, never inferred. All 14 DOIs were checked against the PMID's own PubMed record and resolved at `doi.org` before this issue was built: **0 failures, 0 advisory warnings**. Content derived from PubMed and Crossref metadata; abstracts remain the copyright of their respective publishers.